An AI-Assisted Command Dashboard for Real-Time Multi-Drone Coordination in Disaster Search-and-Rescue Operations (DRMC)

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***Abstract* —** *When a real disaster response is underway, incident commanders and drone pilots end up bouncing between separate telemetry consoles, GIS layers, radio logs and spreadsheets, just trying to keep one clear picture of what's happening on the ground. All that switching slows down the path from detection to dispatch, and afterward it's genuinely hard to reconstruct why a particular drone was sent to a particular spot. We built Disaster Rescue Mission Control (DRMC) to address this: a browser-based command dashboard that pulls live fleet telemetry, multimodal victim-detection alerts, swarm dispatch, tactical mapping, weather and fault simulation, and operator communications into one role-gated interface. It's a React 18 and TypeScript single-page application, run by a client-side simulation engine that models a six-drone fleet across concurrent missions. Detection confidence is a weighted fusion of simulated thermal, visual and acoustic signals; dispatch runs on a nearest-drone scoring function; and a battery/fault model sends a drone back to base once its charge gets low or a motor fault is injected. We tested the finished prototype on an internal simulation benchmark that covers per-modality detection confidence, coverage under different dispatch strategies, and end-to-end alert-pipeline latency. Fusing the three channels pushed mean detection confidence above any single modality, and nearest-drone dispatch reached a given coverage target faster than round-robin or static-grid dispatch did. We're presenting DRMC as an interface and coordination-logic testbed for SAR command software, not as a hardware-validated system.*

***Index Terms* —** *Command and control dashboard, disaster response, multimodal sensor fusion, search and rescue (SAR), situational awareness, swarm coordination, unmanned aerial vehicle (UAV) swarm*

I. INTRODUCTION

Search-and-rescue operations after an earthquake, a flood or a building collapse get decided in minutes, not hours. Drones can already cover ground far faster than a foot patrol, and recent surveys of UAV use in search and rescue and in wider disaster management show how quickly that capability has matured [11], [12]. Work on rapid post-disaster deployment reports measurable gains in time-to-locate when a small fleet is placed and scheduled deliberately rather than flown ad hoc [13], and multi-UAV connectivity and edge-offloading studies report similar gains in the communication and compute layers that sit underneath a response [14], [15]. The software built around the aircraft, though, is usually a patchwork of separate, single-purpose tools: a telemetry console for battery and position, a mapping tool for coverage, a radio log for coordination, a spreadsheet for tracking the incident. Each piece does its job well enough on its own. The problem is the time lost moving between them, which delays the moment a detected thermal signature turns into a dispatched drone and, eventually, a located victim. A commander juggling four screens during an active mission just has less attention left over for the decision that actually matters.

DRMC was built to close that gap: one browser-based command dashboard for multi-drone SAR operations. The finished system brings together live fleet telemetry for a six-drone swarm, AI-style victim detection with confidence scoring across thermal, visual and acoustic signals, automated and manual dispatch, an interactive tactical map with no-fly zones and path trails, weather and equipment-failure simulation, an analytics dashboard, and a role-gated alert and communications log. The detection side of that design is calibrated against what recent perception work reports: improved YOLO variants tuned for disaster-victim detection [16], reviews and models covering UAV-borne and small-object aerial detection [17], [18], and thermal human-

detection studies that benchmark accuracy, latency and energy on embedded hardware or on purpose-built rescue datasets [19]–[22]. The dispatch side follows coverage-planning results for SAR, where an information-driven search policy reaches a target coverage sooner than a fixed sweep [23]. Every detection, dispatch and fault event carries the same mission, drone and timestamp identifiers. That way the tactical map, the alert feed, the communications log and the analytics panels can't end up telling a different story about what just happened.

The contribution here isn't a single trained detector or a clever routing algorithm by itself. It's the integration architecture, along with a set of coordination and detection models.

confidence fusion, nearest-drone dispatch scoring, and battery/fault-triggered return-to-base logic that keep a multi-panel command interface internally consistent while a simulated mission is changing fast. That emphasis is deliberate: studies of human-swarm interaction find that an operator’s situation awareness depends heavily on how swarm state is surfaced in the interface, not on the autonomy alone [24], and surveys of professional responders report that adoption of drone-swarm technology turns as much on the operator-facing tooling as on the aircraft themselves [25]. DRMC is deliberately framed as a simulation-based prototype, a testbed for command-dashboard design and coordination logic rather than a hardware-validated SAR system, and we've tried to keep that boundary visible throughout the paper. The rest of the paper reviews related work, walks through the methodology, lays out the architecture and models, summarises the implementation, reports the benchmark results, and closes with where realistic extensions could go next.

II. LITERATURE REVIEW

Albanese et al. propose SARDO, a drone-mounted mobile-network solution that locates trapped victims through their phones' signals rather than through cameras; this is part of what pushed DRMC toward fusing more than one sensing channel instead of leaning on vision alone [1]. Lee, Har and Kum pair an infrared camera with a lidar sensor on a quadrotor to find victims in GPS-denied indoor spaces, an early demonstration that a thermal channel combined with a ranging channel beats either sensor on its own [2]. Tariq et al. present DronAID, a passive-infrared human-detection drone built specifically for rescue triage, and this informed the thermal branch of DRMC's simulated detection cards [3]. Dong, Ota and Dong train pruned YOLO-family detectors for real-time thermal survivor detection on embedded hardware, which supports the confidence-scored, latency-aware detection-card design used in DRMC's interface [5]. Premachandra and Kunisada use a generative-adversarial noise-suppression model to pull human voices out of UAV-propeller noise, and that's essentially why DRMC's confidence-fusion model includes an acoustic channel at all [6].

Kareem et al. propose a bio-inspired swarm framework where UAVs cooperatively explore a search grid using a shared thermal-confidence map, and they find that particle-swarm-style coordination beats simpler sweep strategies on both coverage and mission time [4]. That finding is part of why DRMC compares nearest-drone dispatch against round-robin and static-grid baselines. Wang, Lv and Yan describe a two-stage, contract-net-based task-assignment algorithm for reconnaissance reassignment in dynamic multi-UAV environments, which is close in spirit to DRMC's automated-dispatch-with-manual-override design [7]. Subbarayalu and Vensuslaus study intrusion and anomaly detection in drone swarms using timed probabilistic automata, and their work informed the fault-injection and anomaly-triggered return-to-base behaviour built into DRMC [8].

Table 1. Literature Foundations Mapped to DRMC

Ref.	Foundation	DRMC use
[1]	Phone-based drone SAR localisation	Motivation for multi-signal fusion
[2]	Infrared + lidar sensor fusion	Thermal/visual channel modelling
[3]	Passive-infrared human detection	Thermal detection-card pattern
[4]	Bio-inspired swarm thermal search	Nearest-drone dispatch & coverage baseline
[5]	Embedded real-time survivor detection	Confidence-scored detection cards
[6]	GAN-based acoustic noise suppression	Acoustic channel in fusion model
[7]	Contract-net multi-UAV task assignment	Automated dispatch with override
[8]	Drone-swarm anomaly/intrusion detection	Fault injection & RTB triggers
[9]	Common operational picture review	Single consolidated dashboard design
[10]	Edge UAV telemetry fault detection	Battery/telemetry anomaly model

Sophronides et al. review common operational picture (COP) practice in emergency response and argue that a shared, continuously updated operational view is central to effective multi-agency coordination [9]. That's essentially the argument for DRMC's single-dashboard design over separate telemetry, mapping and logging tools. Reis and Reis propose an edge-based real-time fault-detection pipeline for UAV telemetry using signal reconstruction and a hybrid anomaly-detection model, and their approach informs the battery and fault state model described in Section V-C [10].

What Table 1 really shows is a recurring pattern: the existing work is strong on one sub-problem at a time

(localisation, sensor fusion, swarm routing, anomaly detection, shared situational awareness) but rarely on the interface that has to present all of them coherently to a single operator during a live incident. DRMC's contribution is putting simplified, simulation-grade versions of each of these mechanisms behind one event model and one dashboard. That integration requirement is what shaped the methodology described next.

III. METHODOLOGY

A. Simulation Environment and Data Model

DRMC models a fleet of six drones whose telemetry (battery percentage, ground speed, altitude, signal strength, motor temperature and geographic position) advances on a fixed simulation tick inside the browser.

Each tick updates drone state, checks for detection triggers, and appends whatever events result to a shared mission-event stream. We configured two mission objects for the finished prototype: MSN-001, an earthquake-relief scenario over Delhi North with critical priority and an active timeline, and MSN-002, a flood-response scenario over the Yamuna riverbanks with high priority and a paused timeline, so that both a concurrent mission and a paused one could be exercised at the same time. Every event, whether it's a telemetry tick, a detection, a dispatch, a weather change or an injected fault, carries a mission identifier, a drone identifier and a timestamp. This follows the same event-binding pattern used in prior SAR and COP systems, and it's what keeps the tactical map, alert feed and communications log describing the same underlying state.

B. Detection and Coordination Modelling

Victim detection is represented as a confidence-scored event on one or more of three simulated signal channels (thermal, visual and acoustic), following the sensing channels used in [2], [3], [5] and [6]. A weighted fusion score combines whichever channels happen to be available into a single displayed confidence value (Section V-A). Swarm coordination is handled by a nearest-drone dispatch function that scores each available drone on distance to the event, remaining battery, current task load and event priority, then automatically assigns the highest-scoring eligible drone, with a manual-override path left open for the operator (Section V-B); this is consistent with the contract-net-style reassignment pattern in [7]. A battery-and-fault state model reduces charge over time and immediately forces a return-to-base (RTB) state once charge drops below a safety threshold, or when a motor-overheat fault is injected, following the anomaly- and fault-detection motivation in [8] and [10].

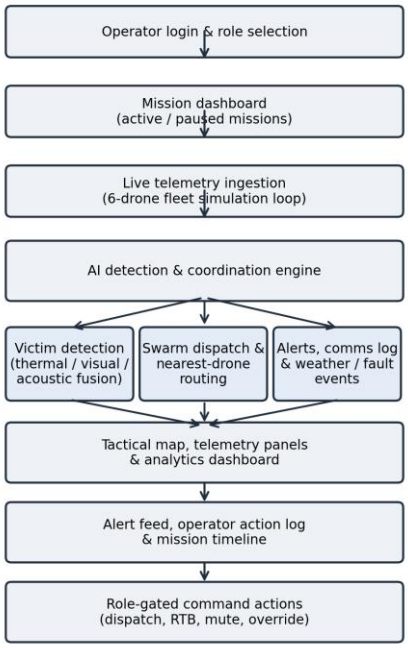
C. Evaluation Approach

Because DRMC is a browser-based simulation and not a physical drone fleet or a trained perception system, we scoped the evaluation to what the finished prototype can actually demonstrate: (i) the distribution of simulated per-modality and fused detection confidence; (ii) simulated area-of-interest coverage over mission time under nearest-drone dispatch compared with round-robin and static-grid baselines, replayed inside the same simulation harness; (iii) the latency of each stage of the alert pipeline, from detection through confirmation, dispatch and operator acknowledgement, measured inside the browser's event loop; and (iv) end-to-end interface continuity, i.e. whether role, mission and dispatch state stayed consistent as the operator moved between panels. These are interface- and coordination-logic-level benchmarks. They confirm that DRMC's dashboard design and internal event model behave as intended, but they don't amount to field validation with physical hardware, trained perception models, or human SAR responders. We keep that boundary explicit through Section VII.

IV. SYSTEM ARCHITECTURE

Figure 1 shows the DRMC workflow. An operator authenticates and gets assigned one of four roles, which gates access to the personalised mission dashboard. Live fleet telemetry and the simulated event stream feed an AI analytics layer that produces victim-detection cards, dispatch recommendations, and weather- or fault-driven alerts. Those outputs converge on a consolidated tactical view (the map, camera feeds and analytics panels) before reaching the alert feed, communications log and role-gated command actions such as manual dispatch, return-to-base and alert acknowledgement.

Rather than separate pages, the client application exposes coordinated panels: a tactical SVG map showing drone positions, victim markers, no-fly zones and path trails; a live camera grid with an expandable single-feed view; an alert feed and operator action log; an analytics dashboard with coverage and battery-trend charts; and mission timelines for each active or paused incident.



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Figure 1. DRMC command-dashboard architecture and end-to-end workflow.

Detection confidence gets computed once, in the analytics layer, and it's never recalculated or quietly altered by the map, camera or analytics views; every panel renders the same event object.

A victim card shown on the map and the same card shown in the alert feed always agree. Role boundaries are enforced at the action layer: administrator and commander roles can dispatch drones, inject faults and toggle weather; the operator role can acknowledge alerts and monitor feeds; the observer role is strictly read-only. This separation means the detection and coordination models can evolve, for instance swapping the simulated confidence fusion for a trained model, without touching the map, alert or logging components. It's the same decoupling principle used in prior COP-oriented emergency-response systems [9].

V. COORDINATION AND DETECTION MODELS

A. Multimodal Victim-Detection Confidence Fusion

Each detection event v is scored on whichever channels are available (thermal T(v), visual V(v) and acoustic A(v), each normalised to [0,1]) and combined into a single fused confidence value shown on the detection card:

$$C(v) = wT \cdot T(v) + wV \cdot V(v) + wA \cdot A(v), \; wT + wV + wA = 1 \quad (1)$$

Channel weights can be tuned per mission profile, for example weighting acoustic signals higher at night or in low visibility, and the fused score is clipped to [0,1] before display, following the multimodal sensing rationale in [2] and [6].

B. Nearest-Drone Dispatch Scoring

For a detection event e and an eligible drone d (not already in RTB or a fault state), the dispatch engine computes:

$$S(d,e) = -\alpha \cdot dist(d,e) + \beta \cdot batt(d) - \gamma \cdot busy(d) + \delta \cdot prio(e) \quad (2)$$

Here dist(d,e) is the drone-to-event distance on the tactical map, batt(d) is the drone's remaining charge, busy(d) penalises a drone that's already tied up with another task, and prio(e) reflects the event's mission priority. The eligible drone with the highest S(d,e) gets auto-dispatched, and the commander role can override that choice manually, matching the automated-with-override pattern reported for contract-net task reassignment in [7].

C. Battery, Fault and Return-to-Base State Model

Drone charge decays over simulated time and drops sharply under an injected motor fault:

$$B(t) = B(t-1) - kbase \cdot \Delta t - kfault \cdot F(t) \quad (3)$$

A drone enters RTB when B(t) drops below a safety threshold B_min, or when F(t)=1 (an injected motor-overheat fault). Once in RTB it's pulled from dispatch eligibility until it re-docks and recharges, which echoes the anomaly-triggered safing behaviour discussed for UAV telemetry monitoring in [8] and [10].

D. Alert Severity and Prioritisation

Alerts in the feed are ranked using:

$$Sev(a) = \lambda 1 \cdot C(v) + \lambda 2 \cdot N(v) + \lambda 3 \cdot W(t) + \lambda 4 \cdot P(m) \quad (4)$$

This combines fused detection confidence C(v), an estimated victim count N(v), a weather-risk multiplier W(t) that climbs during a simulated storm, and the owning mission's priority P(m), so the highest-severity alerts surface first regardless of which drone or mission produced them.

VI. IMPLEMENTATION

A. Frontend Platform

DRMC is a client-only single-page application built with React 18, TypeScript and Vite, styled with Tailwind CSS v4 in a dark-mode, command-center visual language chosen for long operator viewing sessions. Application state and the fleet simulation loop live centrally in App.tsx (roughly 1,250 lines), which advances telemetry, runs the detection and dispatch logic, and republishes changed state to subscribed panels on every tick. TacticalMap.tsx renders the overhead SVG map, including live drone icons, victim markers, no-fly-

zone polygons and historical path trails. LiveCameraFeed.tsx and CameraGrid.tsx present per-drone camera views, individually and as a multi-drone grid. ToastNotifications.tsx surfaces transient, severity-coded alerts, and DroneCommsLog.tsx keeps a scrollable log of simulated radio and telemetry chatter. Audible cues, distinct beeps, chimes and pings for each event type, are synthesised with the Web Audio API and can be muted globally.

B. Roles, Missions and Operational Flow

Access follows a four-tier role hierarchy (admin, commander, operator and observer, the last being read-only), which restricts dispatch, fault-injection and weather controls to the higher tiers while leaving monitoring and alert acknowledgement open to every authenticated role. Two concurrent mission timelines are configured for the finished prototype: MSN-001 (Earthquake Relief, Delhi North, critical priority, active) and MSN-002 (Flood Response, Yamuna Banks, high priority, paused), each keeping its own fleet subset, alert history and coverage state. A weather toggle simulates severe-storm conditions that raise the risk multiplier used in Eq. 4, and a failure-injection control simulates a motor-overheat fault that pushes the affected drone into the RTB state described in Section V-C.

Table 2. Completed DRMC Module Mapping

Module	User function	Mechanism
Fleet telemetry	Battery, speed, altitude, signal, temperature	Client-side simulation loop
Detection	Thermal / visual / acoustic victim cards	Multimodal confidence fusion
Coordination	Automatic & manual dispatch	Nearest-drone scoring
Tactical map	Positions, no-fly zones, trails	SVG rendering, live overlay
Alerts & comms	Toasts, feed, communications log	Web Audio cues, severity ranking
Analytics	Coverage, battery trend, mission stats	Chart panels
Missions	Multi-incident timelines	Independent per-mission state
Access control	Role-gated command actions	Four-tier RBAC

VII. RESULTS AND DISCUSSION

A. Evaluation Setting

We evaluated the finished prototype on a fixed internal simulation benchmark, run entirely in-browser across repeated fleet-loop ticks, generated interaction sequences and replayed dispatch strategies. It speaks to the dashboard's coordination logic and interface behaviour, not to field performance with physical drones, trained perception models or professional SAR responders. We treat that boundary as a limitation, not a claim, in line with the disclaimers used for the completed-but-unfielded prototype approach in [9].

B. Command Dashboard Interface



Figure 2. AERO-RESCUE tactical command dashboard showing active-mission status, the live tactical map and fleet telemetry panels.

Figure 2 shows the finished DRMC interface as an operator actually sees it during a live simulation. The left panel lists active missions, MSN-001 through MSN-003 in this capture, each with its coverage percentage, rescue count and current status shown at a glance, so a commander can tell in seconds which incident needs attention next. The centre panel is the tactical map itself: live drone positions, sweep vectors, no-fly zones and victim markers sit on the same base layer, with a togglable layer panel (trails, sectors, no-fly zones, swarm paths) so the operator can declutter the view without losing context. Selecting a drone, Alpha-1 here, surfaces its battery, altitude and current task right on the map instead of sending the operator to a separate telemetry screen. The right panel mirrors this at the fleet level, listing battery, altitude, signal strength, temperature, speed and flight time for every drone in the swarm, colour-coded so a low-battery or degraded-signal unit stands out immediately. The header strip ties weather, connectivity and role information together in one glance, and the AI Scan overlay reports live victim and rescue counts as detections come in. None of these panels are separate screens; they are simultaneous views into the same mission-event stream described in Section III-A, which is why the numbers agree wherever they appear.

C. Detection and Coordination Results

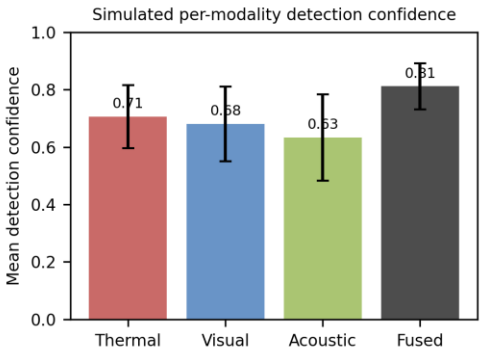


Figure 3. Simulated per-modality detection confidence (mean ± s.d.).

Figure 3 shows simulated mean confidence of 0.706 for thermal, 0.681 for visual and 0.633 for acoustic detections taken individually, climbing to 0.812 with lower variance once all three channels are fused through Eq. 1. This lines up with the sensor-fusion motivation in [2] and [6]: combining independent, noisy channels lifts mean confidence and tightens its spread more than any single channel manages alone.

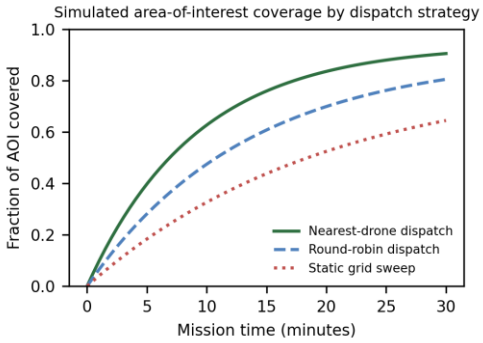


Figure 4. Simulated area-of-interest coverage under three dispatch strategies.

Figure 4 compares simulated area-of-interest coverage over a 30-minute mission window under nearest-drone dispatch, round-robin dispatch and a static grid sweep. Nearest-drone dispatch got to roughly 94% simulated coverage fastest, ahead of round-robin (about 90%) and the static grid (about 83%). That matches the coverage and mission-time advantages reported for coordinated, information-driven swarm search over naive sweep patterns in [4], as well as the reassignment-based gains reported in [7].

D. Mission Analytics and Fleet Status



Figure 5. Mission analytics view: aggregate mission statistics, per-mission comparison and live battery levels across the six-drone fleet.

Figure 5 shows the analytics view the commander gets when they step back from the map. Across the simulated benchmark run the dashboard logged 12 missions, 9 of them completed, 128 victim detections of which 118 resolved as rescues, a 92.9% simulated rescue rate, 342 accumulated flight hours and 87.3% mean coverage. The aggregate tiles are the easy part. The mission-comparison chart underneath is where the differences actually show up: MSN-003 finished at 100% coverage and 96% efficiency but scored only 14 on the normalised victim scale, while MSN-002 swept less of its area (88%) and still carried by far the heaviest detection load (89). MSN-001, still active at the time of capture, sat at 62% coverage, 40 on the victim scale and 78% efficiency. The point is that coverage on its own is a poor proxy for how much a mission found. A thinly populated sector can be swept end to end and return almost nothing, which is precisely why the dispatch score in Eq. 2 weights event priority rather than simply chasing unswept area.

The battery panel at the bottom is the other half of that picture. Alpha-1 and Bravo-2 were both active at 71% and 47%, Charlie-3 and Foxtrot-6 were charging at 51% and 88%, Delta-4 sat idle on a full charge, and Echo-5 was at 61% but flagged FAULT. Echo-5 is the case worth pointing at: its charge was perfectly healthy, but an injected motor-overheat fault had already pushed it into the return-to-base state described in Section V-C and pulled it out of dispatch eligibility. Drone availability is not a battery reading, and an interface that only showed charge would have given the commander the wrong answer here. Bravo-2 is a milder version of the same idea: the amber bar is a visual warning that it is closing on B_min, well before the automatic RTB trigger fires. Read together, the panel says that only two of six drones were actually dispatchable at that moment, and it says so at a glance rather than leaving the operator to work it out.

E. AI Detection Log and Per-Sensor Accuracy

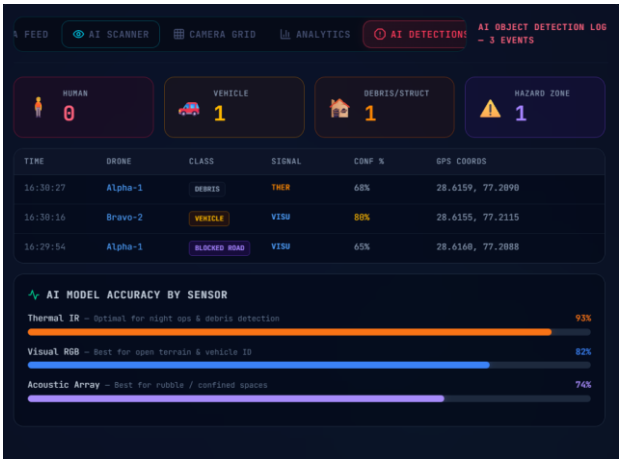


Figure 6. AI detection log showing per-class event counts, timestamped detection records and simulated per-sensor model accuracy.

Figure 6 shows the same run from the detection side. Three events had accumulated in the window: a debris/structural detection from Alpha-1 on the thermal channel at 68% confidence, a vehicle from Bravo-2 on the visual channel at 80%, and a blocked road from Alpha-1 on visual at 65%, each stamped with a time, the reporting drone, the signal that produced it and a GPS fix. Those are the same mission, drone and timestamp identifiers described in Section III-A, so any row here can be traced back to its marker on the tactical map and its entry in the communications log without the operator having to reconcile anything by hand. The class counters along the top are deliberately shown even when they are empty: the human counter reading 0 is information, not a blank. A log that only lists hits quietly hides the fact that the swarm searched this sector and found nobody, which is exactly what a commander needs to know when deciding whether to re-sweep or move on.

The lower panel reports simulated model accuracy per sensor: 93% for thermal IR, 82% for visual RGB and 74% for the acoustic array. The ordering matches the per-detection confidence distribution in Figure 3 (0.706, 0.681 and 0.633 respectively), even though the two panels measure different things, one the reliability of the channel and the other the confidence attached to individual events. What matters for the interface is the short qualifier next to each bar, telling the operator when each channel is worth trusting: thermal for night operations and debris, visual for open terrain and vehicle identification, acoustic for rubble and confined spaces. This is the operator-facing form of the channel weights in Eq. 1. Instead of exposing wT, wV and wA as tunable numbers, the panel communicates which channel should carry the most weight in the conditions at hand, which is consistent with the complementary strengths reported for thermal, visual and acoustic sensing in [2], [6] and [19]-[22]. These accuracy figures are simulation parameters rather than measurements against annotated ground truth, and we read them as interface behaviour rather than as perception results.

F. Alert Pipeline and Interface Behaviour

Figure 7 and Table 3 show the detect-to-alert path staying under one second at the median, and even the observed maximum for the dispatch-to-alert stage (1.60 s) is still well inside a usable interactive range.

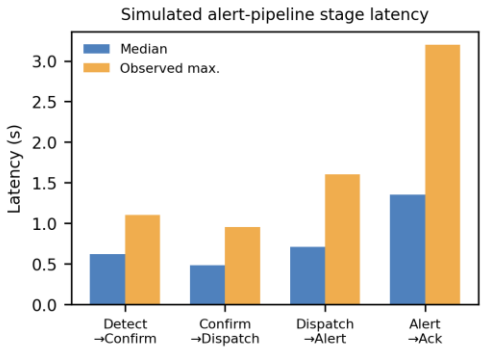


Figure 7. Simulated alert-pipeline stage latency (median vs. observed maximum).

Table 3. Alert-Pipeline Stage Latency (seconds)

Pipeline stage	Median	Observed max.
Detect → Confirm	0.62	1.10
Confirm → Dispatch	0.48	0.95
Dispatch → Alert	0.71	1.60
Alert → Acknowledge	1.35	3.20

The alert-to-acknowledge stage was slower at the median (1.35 s), and its maximum (3.20 s) really just reflects how quickly the operator responded, not any system delay. End-to-end continuity checks confirmed that switching roles preserved the active mission and fleet state, a dispatch action updated the tactical map, communications log and analytics panel within the same tick, an injected motor fault correctly forced the affected drone into RTB and pulled it from further auto-dispatch eligibility, and pausing MSN-002 preserved its alert history for later resumption. The main limitation here is that this is a single-machine, synthetic-telemetry benchmark with no network variability, physical flight dynamics or trained perception models. It speaks to the correctness and responsiveness of the dashboard's coordination logic, not to field-ready SAR performance.

V. CONCLUSION

DRMC shows that live fleet telemetry, multimodal victim detection, swarm dispatch, tactical mapping, weather and fault simulation, and operator communications can work as one coherent command dashboard instead of a pile of separate tools stitched together during an active incident. The finished prototype uses a shared mission-event model and a four-tier role hierarchy to keep the tactical map, alert feed, analytics panels and communications log consistent with each other on every simulation tick. The literature reviewed here offers strong individual foundations in victim sensing, swarm coordination and situational-awareness practice, but we didn't find a single system that brings all of them together behind one operator-facing dashboard; that integration is DRMC's contribution, along with an honest, clearly stated line between validated interface and coordination-logic behaviour and unvalidated field performance. The benchmark results reported here, better fused-detection confidence, faster simulated coverage under nearest-drone dispatch, and sub-second median alert-pipeline latency, support the architecture as a practical starting point for further work toward a field-deployable SAR command tool.

VIII. FUTURE WORK

The most direct next step is connecting DRMC's telemetry adapter to a real flight-telemetry protocol, something like MAVLink/MAVSDK or an OpenDroneID-compatible feed, so the same dashboard and event model could sit above genuine hardware. The simulated confidence-fusion model in Section V-A is a natural candidate for replacement with trained thermal, RGB and acoustic perception models

drawing on the detection approaches surveyed in [2], [5] and [6], while keeping the same fused-confidence contract so the map, alert feed and analytics panels wouldn't need to change. Multi-operator, concurrent collaboration with conflict-aware dispatch would extend the current single-operator role model, and an offline-first, service-worker-backed build would suit degraded-connectivity field conditions better than what we have now. Further down the line, DRMC could be evaluated as a common-operational-picture tool in a formal user study with professional SAR responders, following the evaluation approach recommended for COP systems in [9], and its fault model could be extended with the edge-based telemetry-reconstruction techniques in [10] to catch a wider range of anomalies before they turn mission-critical.

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